

Aaditya Shrivastava

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SUMMARY

Mechanical and Robotics Engineer with an MEng from UC Berkeley focused on robot hardware, electromechanical integration, and physics-based simulation. Built robotic systems end-to-end at IIT Bombay and UC Berkeley including soft robotic arm and a wheeled mobile robot for hospitals. Additional depth in regulated aerospace manufacturing (First Article Inspection, non-conformance issuance, CMM inspection, 8D root cause) and physics-based digital twin development validated against real sensor data.

EDUCATION

University of California, Berkeley, CA, USA

May 2026

Master of Engineering in Mechanical Engineering, Control of Robotic and Autonomous Systems Concentration

GPA: 3.60/4.00 · Courses: Robotic Locomotion, Model Predictive Control, Design of Microprocessor-based Mechanical Systems, Modelling, Simulation and Digital Twins

K.J. Somaiya College of Engineering, Somaiya Vidyavihar University, Mumbai, India

Jun 2025

Bachelor of Technology in Mechanical Engineering

GPA: 9.64/10.00 · Courses: Finite Element Modelling, Computer Aided Engineering, Mechatronics, Theory of Design, Automation

TECHNICAL SKILLS

- Mechanical Design and CAD: SolidWorks (CSWA and CSWPA-SM certified), Creo; multi-DOF mechanism and end effector design; actuator sizing and selection (cycloidal, strain-wave/harmonic, planetary); GD&T to ASME Y14.5; tolerance analysis; DFM/DFA; FEA (Ansys Workbench) for structural, thermal, and fatigue analysis; multibody dynamics (MSC Adams); KiSSsoft gear analysis
- Electromechanical Integration: PCB design (KiCad); HV/LV wire harness design and routing; motor control and driver integration; sensor integration (cameras, encoders, IMUs, force sensors); embedded systems (ESP32, Raspberry Pi); inter-device communication (SPI, I2C, UART, CAN); Hardware-in-the-Loop (HIL) testing; oscilloscope and logic analyzer debugging
- Simulation and Digital Twins: Physics-based modelling (lumped-parameter, Lagrangian dynamics, ODE systems); model validation against real sensor data (RMSE, MAE); Genetic Algorithm optimization; surrogate ML models; Gym-style RL environment design; multi-agent swarm simulation; SOFA FEM for deformable objects; NVIDIA Isaac Sim fundamentals
- Manufacturing and Quality: AS9100 regulated manufacturing; APQP, PPAP; First Article Inspection (full, partial, delta per AS9102); non-conformance issuance and 8D root cause investigation; GD&T audits per ASME Y14.5; CMM (FARO Arm, Pininfarina CMM); laser QC; XRF alloy verification; hardness testing (Rockwell, Brinell, Vickers); Kaizen and LEAN
- Programming and Tools: Python (NumPy, SciPy, PyTorch, Gymnasium, pandas); MATLAB/Simulink; Bash; Linux; Git and GitHub; SAP; SQL; LabVIEW

ROBOTIC HARDWARE DESIGN AND INTEGRATION

ARMS Lab, IIT Bombay - Systems and Controls Engineering Dept - Dr. Arpita Sinha, Mumbai, India

Jan - Jul 2025

Soft Robotics Research Intern

- Designed and built a five-segment cable/tendon-driven robotic manipulator from concept through validated hardware for an industry-sponsored autonomous refueling project: solved the S-shape vs C-shape bending problem by designing intermediate cable anchor points enabling independent actuation of proximal and distal segments; sized actuators, integrated motor controllers, encoders, force sensors, and embedded electronics
- Ran FEA on compliant segments under cyclic loading in parallel with physical iterations; selected materials for tendons and structures based on strength, friction, and durability; built a granular gripper evaluating soft fill materials (ground coffee, dried pulses) for grip and jamming characteristics; developed a neural network forward kinematics model in Python validated against Qualisys motion-capture ground truth; collaborated with a computer vision team on autonomous task execution

Wheeled Mobile Robots for Hospitals - UC Berkeley Capstone, Prof. H. Kazerooni

Sept 2025 - May 2026

- Owned the full mechatronics stack of a last-mile delivery robot: SolidWorks assemblies and GD&T drawings; custom PCB design (KiCad); HV/LV wiring architecture; motor driver, encoder, and camera integration; embedded firmware; system-level bring-up and validation; identified and resolved cross-system failures across mechanical, electrical, and firmware interfaces during commissioning

SIMULATION AND DIGITAL TWIN DEVELOPMENT

Data Center Thermal Digital Twin and RL Environment - Independent Project

Mar - Apr 2026

github.com/t-rexx1/NovaCool_DigiTwin

- Built a physics-based thermal model of a 10MW air-cooled data center: diagnosed physically impossible 132°C output using first-principles thermodynamics, back-calculated correct airflow parameters from sensor data, and validated to RMSE 4.09°C across 288,000 data points; built a Gym-style RL environment (13D observation, 8D action) with a dead-band heuristic baseline achieving 3.0% PUE improvement

github.com/t-rexx1/Aerial-Firefighting-Swarm-Drones-

- Built physics-based digital twins of drone swarms, satellite constellations, and aerial firefighting missions in Python; applied Genetic Algorithm optimization to evolve control policies, coverage patterns, and path planning; implemented surrogate ML models reducing full simulation evaluation calls by 60% while maintaining optimization accuracy

MANUFACTURING AND QUALITY

Mahindra Aerospace, Bangalore, India

Jun - Aug 2024

Industrialization Intern

- Worked on AS9100 regulated manufacturing floor producing precision sheet metal aerospace components for Boeing and Airbus across the full process cycle (CNC routing, laser cutting, CNC machining on 3/4/5 axis, deburring, press brake forming, hydraulic press); operated CMM (FARO Arm, Pininfarina CMM), laser QC, XRF gun, and hardness testers (Rockwell, Brinell, Vickers); executed First Article Inspections (full, partial, delta per AS9102)
- Personally issued non-conformances against out-of-tolerance press tooling at a sub-tier supplier during on-site quality audit; led 8D root cause investigations through verified closure; performed GD&T audits per ASME Y14.5; APQP documentation; contributed to Kaizen and LEAN continuous improvement

SIMULATION AND CONSULTING

Tata Advanced Systems Ltd, Mumbai, India

Jun - Jul 2023

Mechanical Design Intern

- Built MSC Adams + Simulink co-simulation of a robotic loading arm; applied kinematic and structural analysis to evaluate actuator loads and stress distribution across strain-wave and planetary gear trains before physical build; KiSSsoft gear specification for tooth geometry and load distribution

SCAILAB - Deep Tech Entrepreneurship Course, UC Berkeley

Jan - May 2026

Robotics Technology Consultant

- Technical consulting engagement with a Swedish deep tech robotics startup: evaluated the full hardware and software tech stack against real-world deployment requirements; contributed technical perspective on simulation and autonomy stack feasibility for investor materials; conducted customer discovery interviews; presented findings to 100+ investors and industry partners at UC Berkeley Demo Day

CERTIFICATIONS

- CSWA and CSWPA-SM - Dassault Systèmes (SolidWorks Professional Certifications, including Sheet Metal)
- Wheeled Mobile Robots, Experimental Robotics (Coursera)
- Robotics Engineering and Applications (L&T Edutech)
- Joy of Computing using Python (NPTEL)